Exam :	Quiz 2
Subject :	Deep Learning
Total Marks :	40.00
QP :	2025 Mar16: IIT M AN EXAM QIP4

Exam Mode | Learning Mode

View Question Paper Summary

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QUESTION MENU

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0.00 / 40.00

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Question 1 : 6406531175175 Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT **"DEGREE LEVEL : DEEP LEARNING (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)**

OPTIONS :

- YES
- NO

Your score : 0

Discussions (0) Suggest Correction Broken Format?

Question 2 : 6406531175176

View Solutions (1) Total Mark : 3.00 | Type : MCQ

What is the impact of using a sigmoid activation function on the gradients when the input to the neuron is a very large negative number?

OPTIONS :

- The output approaches 0 and the gradient approaches 0.
- The output approaches 0.5 and the gradient approaches 0.25.
- The output approaches 1 and the gradient approaches 0.
- The output approaches 0 and the gradient approaches 1.

Your score : 0

Discussions (0) Suggest Correction Broken Format?

Question 3 : 6406531175177

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Why is bias correction needed in the Adam optimizer?

OPTIONS :

- To prevent the learning rate from becoming too large in the early iterations.
- To correct the underestimated moving averages of the first and secondmoments at the beginning of training.
- To increase the momentum effect for faster convergence.
- To ensure the learning rate remains constant throughout training.

Your score : 0

Discussions (0) Suggest Correction Broken Format?

Question 4 : 6406531175178

View Solutions (0) Total Mark : 3.00 | Type : MCQ

How do skip connections in ResNet work?

OPTIONS :

- By incorporating element-wise addition of the input of a layer to its output
- By concatenating the input and output of a layer
- By multiplying the input and output of a layer
- By removing the input from the output

Your score : 0

Discussions (0) Suggest Correction Broken Format?

Question 5 : 6406531175179

View Solutions (0) Total Mark : 3.00 | Type : MCQ

What is the primary purpose of L2 regularization in machine learning models?

OPTIONS :

- To prevent overfitting by adding the sum of absolute values of weights to theloss function
- To encourage sparsity in the model by setting some weights to zero
- To prevent overfitting by adding the sum of squared values of weights to theloss function
- To increase the complexity of the model by penalizing small weights

Your score : 0

Discussions (0) | Suggest Correction | Broken Format?

Question 6 : 6406531175180

View Solutions (0) Total Mark : 3.00 | Type : MCQ

What is the soft target used when injecting noise at the output?

OPTIONS :

- A label that is either 0 or 1, with no probability distribution
- A target value that is continuous or probabilistic rather than discrete
- A random label assigned to misclassified samples
- A label that is always equal to the predicted output

Your score : 0

Discussions (0) | Suggest Correction | Broken Format?

Question 7 : 6406531175181

View Solutions (0) Total Mark : 3.00 | Type : SA

Consider a neural network architecture for a regression task. Three distinct subsets of the training dataset are used to train three distinct models: A,B and C. Note that these models have the same architecture with different parameter values. Consider a test dataset of size four with true labels: y = [1,3,6,8]. The predictions of the models on this dataset is given below:
• Model A predictions: [1, 4, 7, 8] • Model B predictions: [3, 5, 6, 9] • Model C predictions: [2, 3, 5, 7]
Calculate the variance of these models averaged over all data-points in the test dataset. Enter your answer up to 2 decimal places.

Answer (Numeric):

Answer

Accepted Answer : 0.62 to 0.72

Your score : 0

Discussions (0) | Suggest Correction | Broken Format?

Question 8 : 6406531175182

View Solutions (0) Total Mark : 3.00 | Type : SA

The weighted moving average (WMA) is used to smooth data by giving more importance to recent values. Suppose we compute a weighted moving average for a sequence of gradient values using the formula:

$$v_t = \beta v_{t-1} + (1-\beta)\nabla w_t$$

where,

- v_t is the weighted moving average at time step t.
- $\beta = 0.9$ is the decay factor.
- ∇w_t is the gradient value at time step t.
- $v_{-1} = 0$

If the gradient values at $t = 0, 1, 2$ are $\nabla w_0 = 5, \nabla w_1 = 3$ and $\nabla w_2 = 4$. Compute v_2. Enter your answer correct up to 3 decimal places.

Answer (Numeric):

Answer

Accepted Answer : 1.065 to 1.085

Your score : 0

Discussions (0) Suggest Correction

Question 9 : 6406531175183

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Given an input array X and a kernel/filter K as follows:

$$X = \begin{bmatrix} 1 & 0 & -2 & 1 \\ 1 & 2 & 0 & 1 \\ -1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Your score : 0

Broken Format?

Question 10 : 6406531175184

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : MCQ

Convolve the kernel K over the input X with a stride $s = 1$ and no padding to obtain matrix A. Which of the following options represents matrix A?

OPTIONS :

○ $\begin{bmatrix} -1 & 1 \\ 2 & 5 \end{bmatrix}$

- $\begin{bmatrix} \frac{1}{9} & \frac{1}{9} \\ \frac{2}{9} & \frac{5}{9} \end{bmatrix}$
- 9
- $\begin{bmatrix} -1 & -1 \\ -2 & -5 \end{bmatrix}$

Your score : 0

Discussions (0) Suggest Correction

Question 11 : 6406531175185

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : MCQ

Apply average pooling of size 2×2 on A to produce matrix B. Which of the following options represents matrix B?

OPTIONS :

- $\begin{bmatrix} -1 & 1 \\ 2 & 5 \end{bmatrix}$
- $\begin{bmatrix} \frac{-1}{9} & \frac{1}{9} \\ \frac{2}{9} & \frac{5}{9} \end{bmatrix}$
- 1.75
- $\begin{bmatrix} \frac{1}{9} & \frac{-1}{9} \\ \frac{-2}{9} & \frac{-5}{9} \end{bmatrix}$

Your score : 0

Discussions (0) Suggest Correction

Question 12 : 6406531175186

View Parent QN View Solutions (0) Total Mark : 3.00 | Type : SA

Pass B through the sigmoid (logistic) function with weight equal to 1 and no bias to get the final output $\hat{y}$.
L is some loss function that depends on $\hat{y}$. Given that $\frac{\partial L}{\partial \hat{y}} = -1$, determine the value of $\frac{\partial L}{\partial K_{00}}$ where K_{00} is the element of K at index $(0, 0)$. Note that we use zero-indexing here.
Enter your answer correct up to 3 digits after the decimal.

Answer (Numeric):

Answer

Accepted Answer : -0.136 to -0.116

Your score : 0

Discussions (0) Suggest Correction

Question 13 : 6406531175187

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider a loss function, $L(w) = w^3 + 2w$, where w represents the model parameter. The AdaGrad optimizer is used to minimize this loss. The learning rate η is set to 0.5, and at the initial iteration ($t = 0$), the parameter w has a starting value of $w_0 = 2$.

AdaGrad Update Rule:

$$v_t = v_{t-1} + (\nabla w_t)^2$$
$$w_{t+1} = w_t - \frac{\eta}{\sqrt{v_t + \epsilon}} * \nabla w_t$$
$$v_{-1} = 0$$
$$\text{use } \epsilon = 0$$

Enter your answer correct up to two decimal places for all sub-questions.
For intermediate calculations, use three decimal places so that your final answer falls within the expected range.

Your score : 0

Broken Format?

Question 14 : 6406531175188

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What will be the value of the parameter w after the first iteration (i.e. w_1 when t=1)?

Answer (Numeric):

Answer

Accepted Answer : 1.4 to 1.6

Your score : 0

Discussions (0) Suggest Correction

Question 15 : 6406531175189

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What will be the value of the accumulated sum of squared gradients v after the first iteration (i.e. v_1 when t=1)?

Answer (Numeric):

Answer

Accepted Answer : 272.46 to 272.66

Your score : 0

Discussions (0) Suggest Correction

Question 16 : 6406531175190

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What will be the value of the gradient w after the first iteration (i.e. ∇w_1 when t=1)?

Answer (Numeric):

Answer

Accepted Answer : 8.60 to 8.90

Your score : 0

Discussions (0) Suggest Correction

Question 17 : 6406531175191

View Parent QN View Solutions (0) Total Mark : 2.00 | Type : SA

What will be the value of the parameter w after the second iteration (i.e. w_2 when t=2)?

Answer (Numeric):

Answer

Accepted Answer : 1.08 to 1.38

Your score : 0

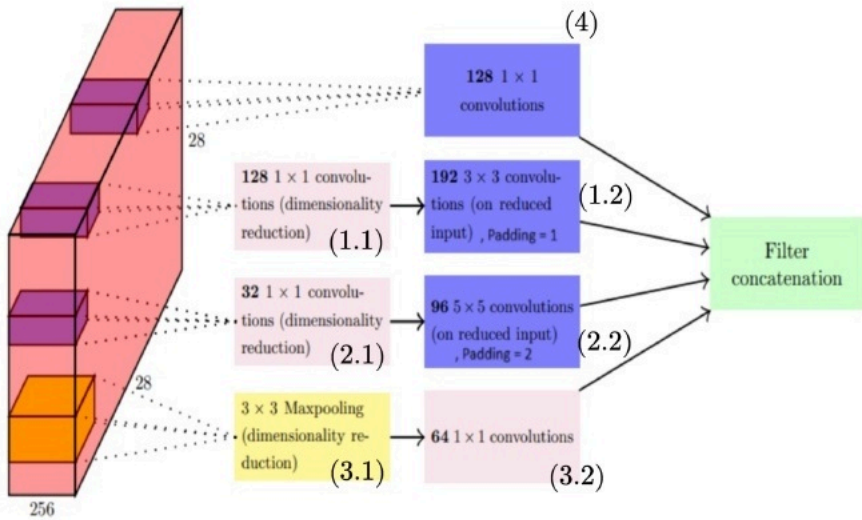
Discussions (0) Suggest Correction

Question 18 : 6406531175192

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the inception module used in the GoogLeNet architecture for input size $28 \times 28 \times 256$. There are four blocks numbered **(1)**, **(2)**, **(3)**, **(4)**. Blocks **(1)**, **(2)**, **(3)** are made up of two layers each, denoted as **(x.1)** and **(x.2)** in that sequence.



- input = 28 x 28 x 256
- F = Spatial Size
- K = Number of filters
- S = Stride = 1
- P = Padding size (zero-padding size)

Suppose Width, Height and Depth (number of channels) are denoted by W, H, D respectively, then the **feature map volume** of the tensor (3-D array) $= W \times H \times D$ (multiplication of W, H, D). The sub-questions will go through the blocks in this sequence: $(4) \rightarrow (1) \rightarrow (2) \rightarrow (3)$.

Your score : 0

Broken Format?

Question 19 : 6406531175193

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What is the feature map volume after applying **(4)**, which is the convolution layer with $F = 1, K = 128, S = 1$ and $P = 0$, on the input?

Answer (Numeric):

Answer

Accepted Answer : 100352

Your score : 0

Discussions (0) Suggest Correction

Question 20 : 6406531175194

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What is the total number of parameters in (4), which is the convolution layer with $F = 1$, $K = 128, S = 1$ and $P = 0$? Ignore the biases.

Answer (Numeric):

Answer

Accepted Answer : 32768

Your score : 0

Discussions (0) Suggest Correction

Question 21 : 6406531175195

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What is the feature map volume after applying the two convolution layers, (1.1) and (1.2), in succession on the input?

- (1.1): $F = 1, K = 128, S = 1$ and $P = 0$
- (1.2): $F = 3, K = 192, S = 1$ and $P = 1$

Answer (Numeric):

Answer

Accepted Answer : 150528

Your score : 0

Discussions (0) Suggest Correction

Question 22 : 6406531175196

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What is the total number of parameters in the two convolution layers, (1.1) and (1.2), which were applied in succession on the input? Ignore the biases.

- (1.1): $F = 1, K = 128, S = 1$ and $P = 0$
- (1.2): $F = 3, K = 192, S = 1$ and $P = 1$

Answer (Numeric):

Answer

Accepted Answer : 253952

Your score : 0

Discussions (0) Suggest Correction

Question 23 : 6406531175197

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : MCQ

What is the feature map dimensions, $(W \times H \times D)$, after applying the two convolution layers, **(2.1)** and **(2.2)**, in succession on the input?

- (2.1): $F = 1, K = 32, S = 1$ and $P = 0$
- (2.2): $F = 5, K = 96, S = 1$ and $P = 2$

OPTIONS :

- 28 X 28 X 96
- 24 X 24 X 96
- 28 X 28 X 128
- 32 X 32 X 128

Your score : 0

Discussions (0) Suggest Correction

Question 24 : 6406531175198

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What is the total number of parameters in the two convolution layers, **(2.1)** and **(2.2)**, which were applied in succession on the input? Ignore the biases.

- (2.1): $F = 1, K = 32, S = 1$ and $P = 0$
- (2.2): $F = 5, K = 96, S = 1$ and $P = 2$

Answer (Numeric):

Answer

Accepted Answer : 84992

Your score : 0

Discussions (0) Suggest Correction

Question 25 : 6406531175199

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : MCQ

Which of the following option represents the feature map dimensions $(W \times H \times D)$ after applying **(3.1)**, the MaxPooling layer with $F = 3, S = 1$ and $P = 1$ on the input?

OPTIONS :

- 3 X 3 X 256
- 28 X 28 X 1
- 26 X 26 X 64
- 28 X 28 X 256
- Insufficient Information

Your score : 0

Discussions (0) Suggest Correction

Question 26 : 6406531175200

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : SA

What is the total number of parameters in (3.1), the Maxpooling layer with $F = 3$, $S = 1$ and $P = 1$?

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0) Suggest Correction

Question 27 : 6406531175201

View Parent QN View Solutions (0) Total Mark : 1.00 | Type : MCQ

Which of the following option represents the feature map dimensions ($W \times H \times D$) after concatenating all the feature maps that come out of the four blocks?

OPTIONS :

- 28 X 28 X 192
- 28 X 28 X 256
- 28 X 28 X 480
- Insufficient Information

Your score : 0

Discussions (0) Suggest Correction

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